

Teaching Exam

Subject – Computer Science

Topic – Computer Network



Lecture No.- 03

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Topics to be Covered



Topic

- 1:- Network Device ✓
- 2:- Types of Network and its application ✓
- 3:- Advantage and Disadvantage of Network ✓
- 4:- Network Topology and its types , Application, Advantage and Disadvantage. ✓
- 5:- Importance and Conclusion of Network Topology ✓

Topic- Network Device – GATEWAY

What is Gateway?

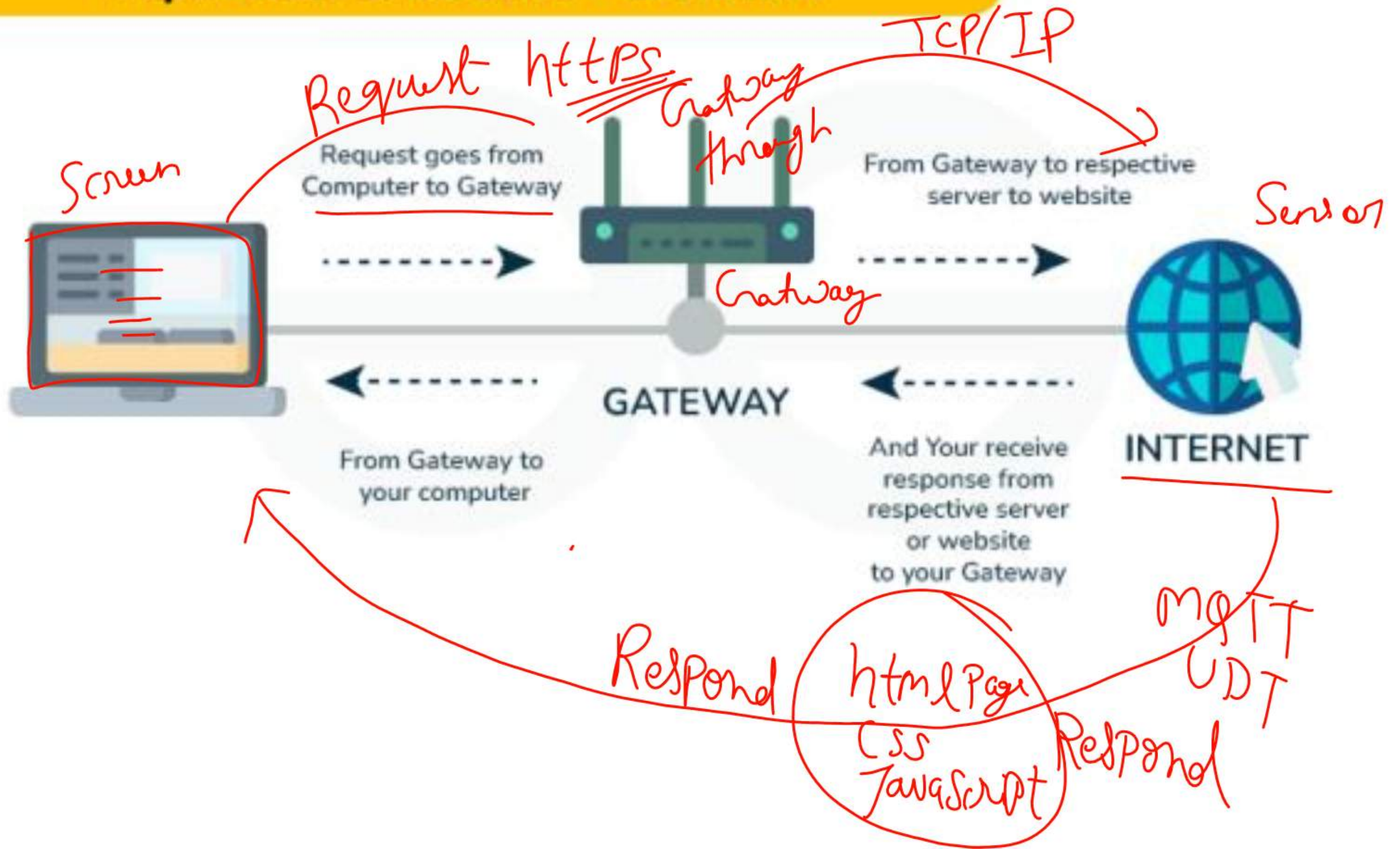
A gateway is a connecting point of any network that helps it to connect with different networks. The gateway monitors and controls all the incoming and outgoing traffic of the network. Suppose there are two different networks and they want to communicate with each other, so they need to set up a path between them. Now that path will be made between gateways of those different networks. Gateways are also known as protocol converters because they help to convert protocol supported by traffic of the different networks into that are supported by this network. Because of that, it makes smooth communication between two different networks.

How does Gateway Work?

Gateway has a simple working methodology of five steps:

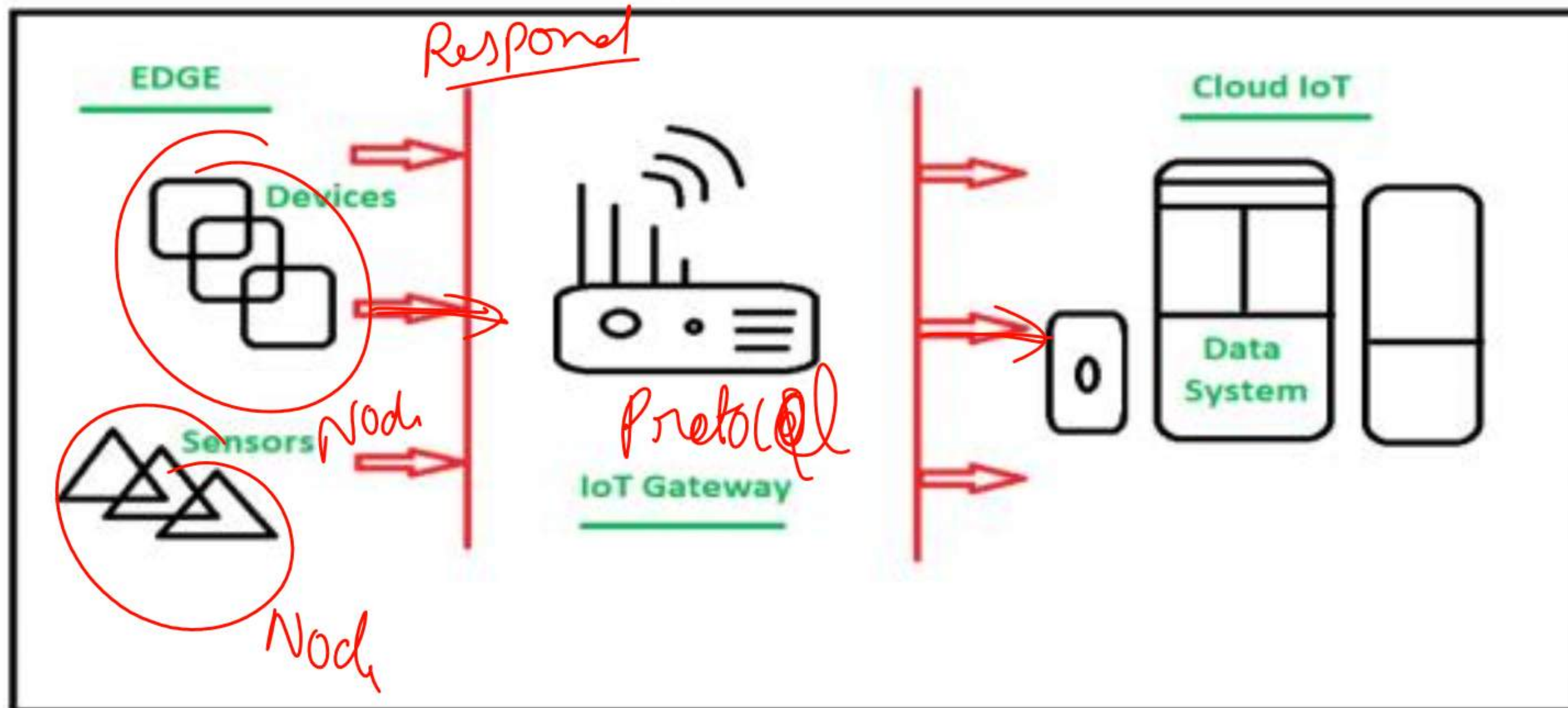
- **Step 1:** It gets data from the network
- **Step 2:** It intercepts and analyzes the received data.
- **Step 3:** It routes the data to the destination address.
- **Step 4:** It converts the received data to make that compatible with the receiver network.
- **Step 5:** It sends the final data inside the network.

Topic- Network Device – GATEWAY



Topic- Network Device – GATEWAY

Gateway is a device that connects different networks. In simpler terms, it is like a bridge. Its function is to connect the device in one network to interact with a device in another network whereas



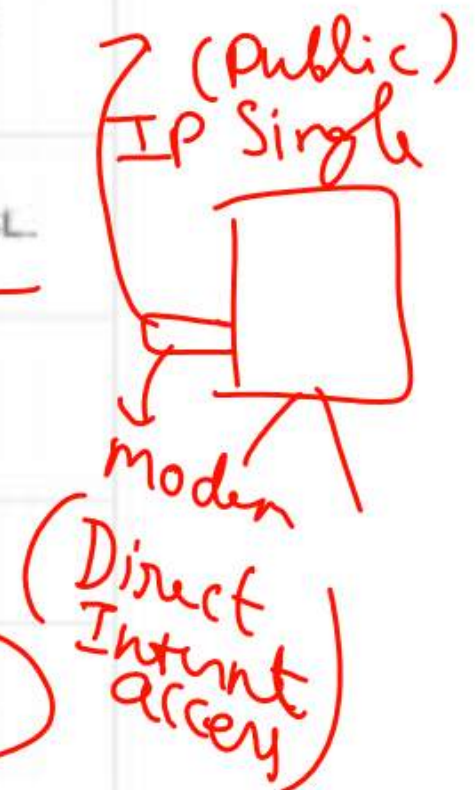
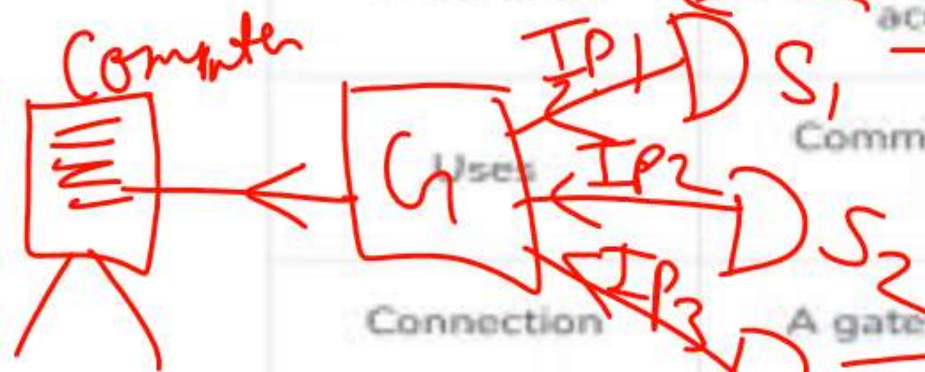
Topic- Network Device – Difference between GATEWAY & ROUTER

Difference between Gateways and Router

 Gateways	 Router
A gateway is a device that is used for communication between networks having different sets of protocols.	Route is a device that receives, analyzes, and forwards the data packets to other networks.
Gateway connects two different networks.	It routes the data packets via the same networks.
Gateway does not support dynamic routing.	Router supports the dynamic routing.
The main function of a gateway is protocol translation.	The main function of a router is routing the traffic from one network to another network.
It is hosted on dedicated applications, physical servers, or virtual applications.	It is hosted on only the dedicated applications.
The gateway is also called a gateway router, proxy server, and voice gateway.	The router is also called a wireless router and an Internet router.

Topic- Network Device – Difference between GATEWAY & MODEM

Basis of Comparison	Gateway	Modem
Definition	It is a device that is the center point for two or more networks to communicate with each other.	A device that connects a home or office device to the internet.
Function	Any device that enables traffic flow in and out of the network can function as a gateway.	It converts the digital signals to analog to transmit them in cable lines later converting them back to digital at the receiver.
Integration	It includes a <u>router that directs traffic between devices on a network.</u>	Does not include a router instead mainly on the connection of devices to the internet.
Protocols	May include protocols like a <u>firewall, Wi-Fi access point, or Ethernet switch.</u>	<u>Supports one or two protocols such as cable or DSL.</u>
Uses	<u>Commonly used in businesses and large organizations.</u>	<u>Commonly used in homes.</u>
Connection	<u>A gateway connects to multiple devices.</u>	<u>A modem connects to a single device.</u>
IP address management	<u>A gateway can support multiple private IP addresses of multiple devices</u>	<u>A modem supports a single public IP address.</u>
Cost	<u>Gateway is expensive.</u>	<u>The modem is cheaper than Gateway.</u>

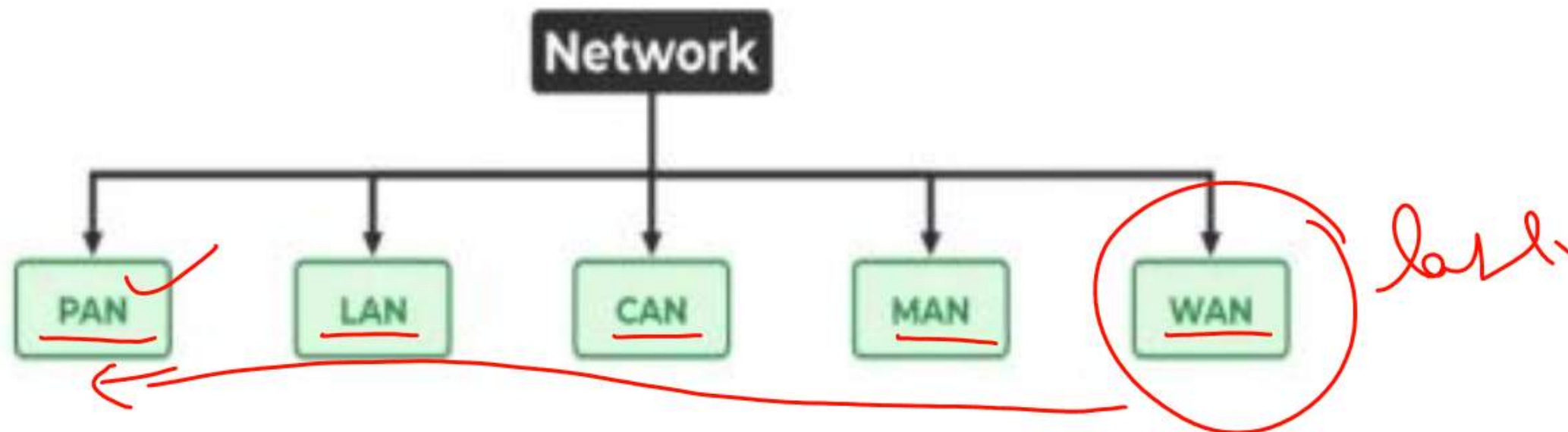


Types of Computer Networks

Computer networks are classified based on several factors such as geographical area, ownership, architecture, topology, and transmission technology. The below is the classification of the networks :

Classification Based on Geographical Area

This is the most common way of classifying networks below are the different types :



Range Based
Network Type

5 Renter
wime

Topic- Types of Computer Network-PAN

1. Personal Area Network (PAN)

Communication medium

wired
USB

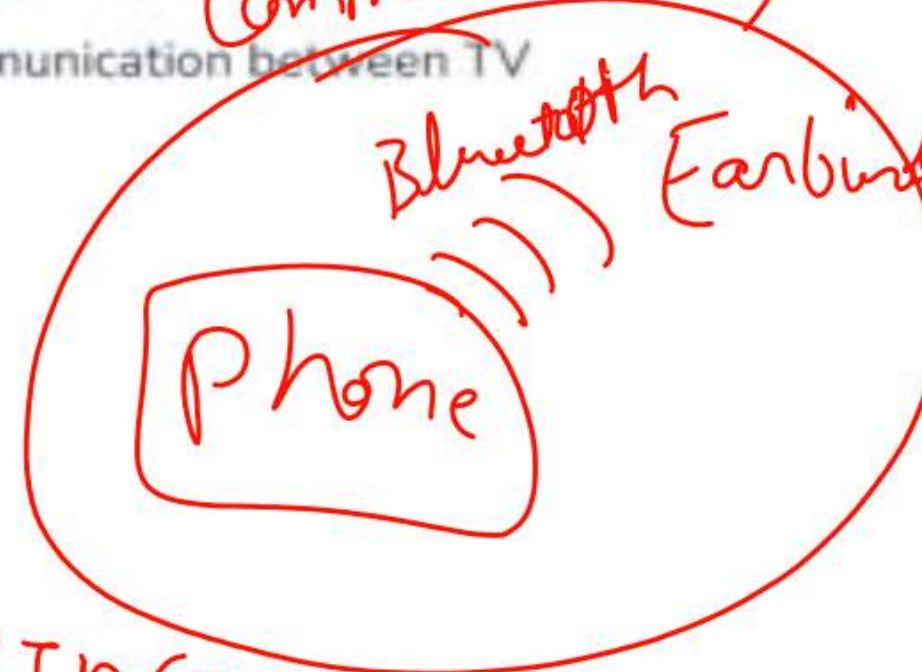
wireless

Bluetooth

RFID

NFC

(Near field Communication)



Bluetooth, USB Cable



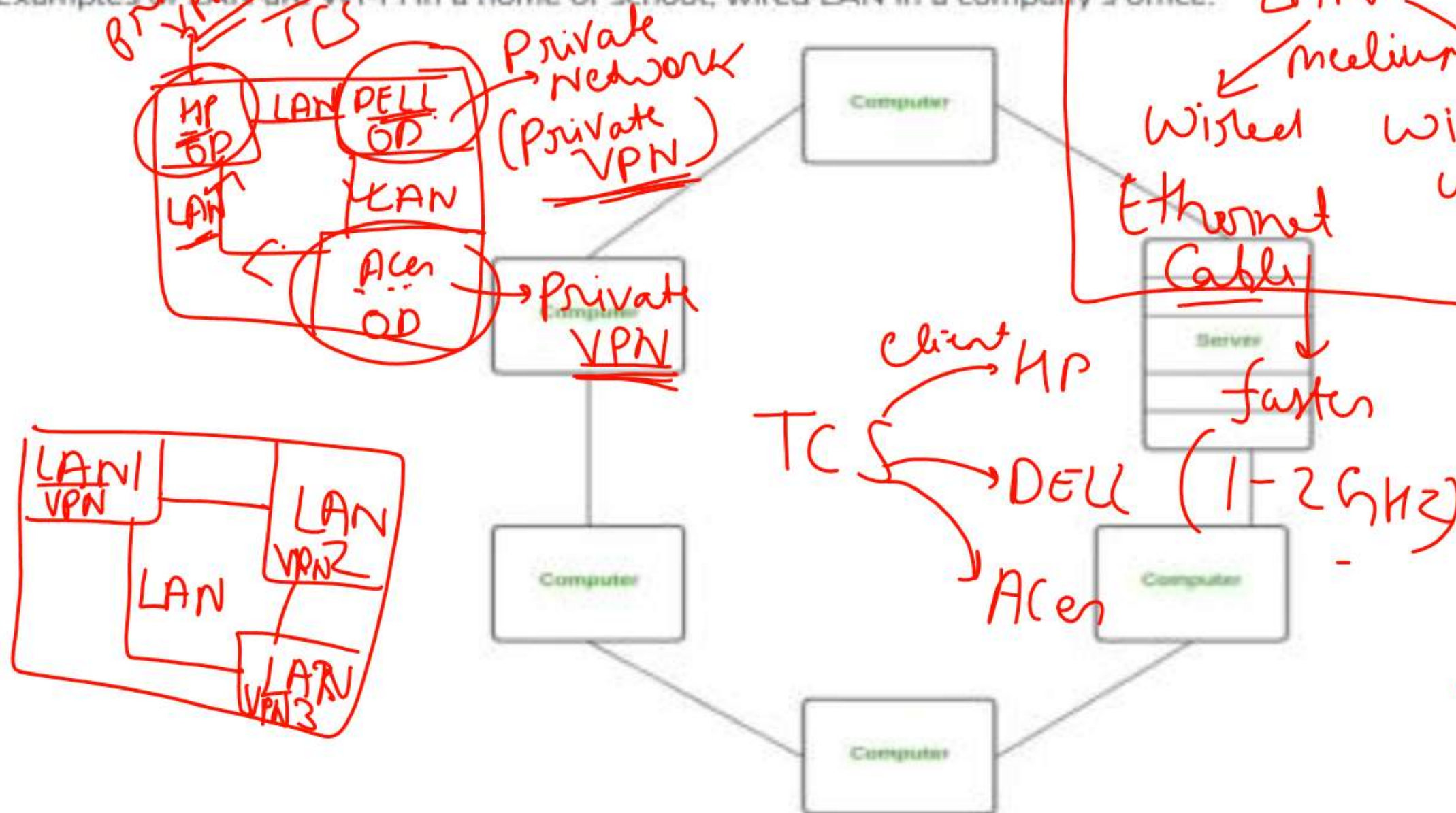
RFID (Radio frequency Identification)
RFID to Barcode
RFID Reader
Reflecti

Topic- Types of Computer Network-LAN

2. Local Area Network (LAN)

LAN is the most frequently used network. It is a computer network that connects computers through a common communication path, contained within a limited area, that is, locally. A LAN encompasses two or more computers connected over a server. The two important technologies involved in this network are Ethernet and Wi-Fi. It ranges up to 2km & transmission speed is very high with easy maintenance and low cost.

Examples of LAN are Wi-Fi in a home or school, wired LAN in a company's office.



Local Area Network

CMD
ping google.com
Replies
16ms
17ms
20ms

Common Network

Within floor
On building

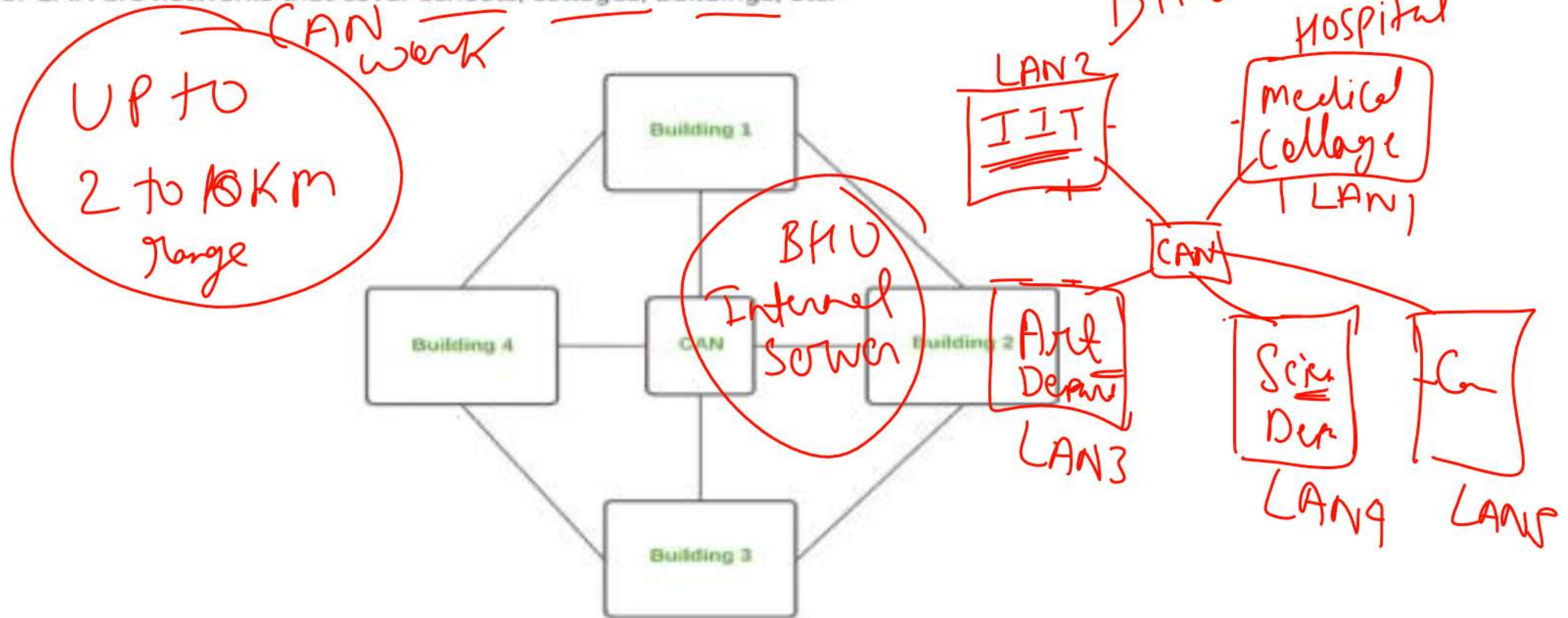
Range up to 2km (500m - 1km)

Topic- Types of Computer Network-CAN

3. Campus Area Network (CAN)

CAN is bigger than a LAN but smaller than a MAN. This is a type of computer network that is usually used in places like a school or colleges. This network covers a limited geographical area that is, it spreads across several buildings within the campus. CAN mainly use Ethernet technology with a range of few kilometers. Its transmission speed is very high with a moderate maintenance cost and moderate cost.

Examples of CAN are networks that cover schools, colleges, buildings, etc.



Topic- Types of Computer Network-MAN

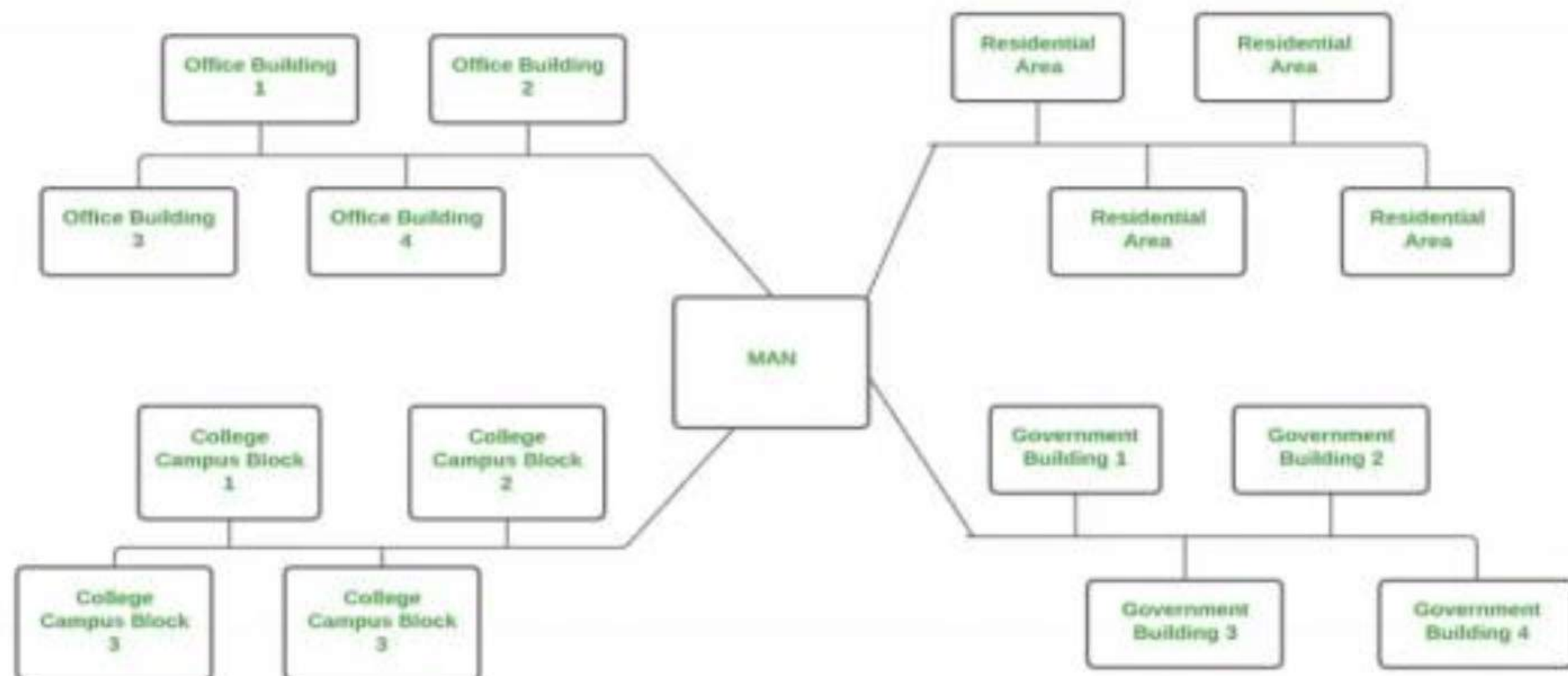


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4. Metropolitan Area Network (MAN)

A MAN is larger than a LAN but smaller than a WAN. This is the type of computer network that connects computers over a geographical distance through a shared communication path over a city, town, or metropolitan area. This network mainly uses FDDI, CDDI, and ATM as the technology with a range from 5km to 50km. Its transmission speed is average. It is difficult to maintain and it comes with a high cost.

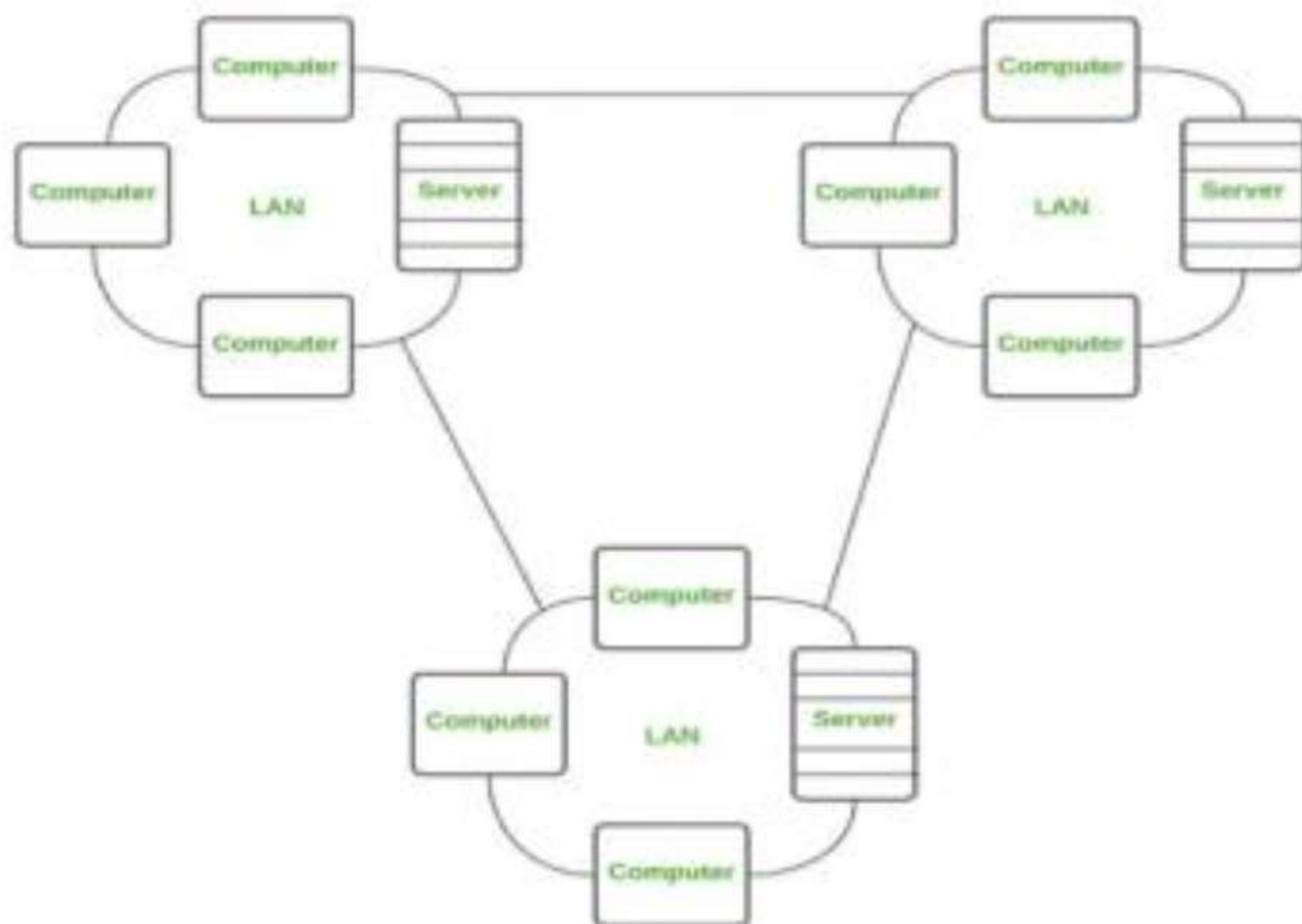
Examples of MAN are networking in towns, cities, a single large city, a large area within multiple buildings, etc.



5. Wide Area Network (WAN)

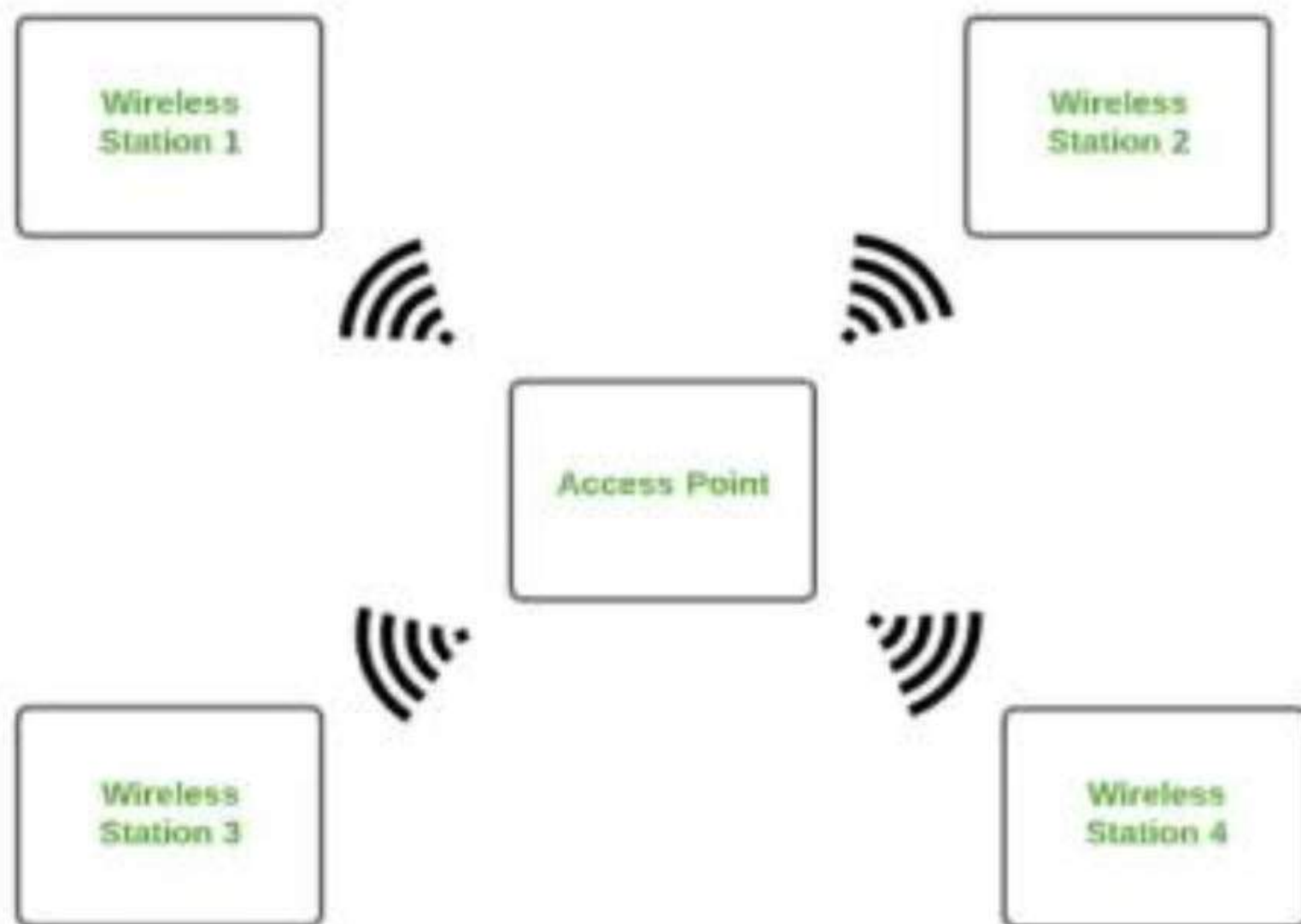
WAN is a type of computer network that connects computers over a large geographical distance through a shared communication path. It is not restrained to a single location but extends over many locations. WAN can also be defined as a group of local area networks that communicate with each other with a range above 50km. Here we use Leased-Line & Dial-up technology. Its transmission speed is very low and it comes with very high maintenance and very high cost.

Examples of WAN are the Internet (largest WAN), Banking networks linking global branches



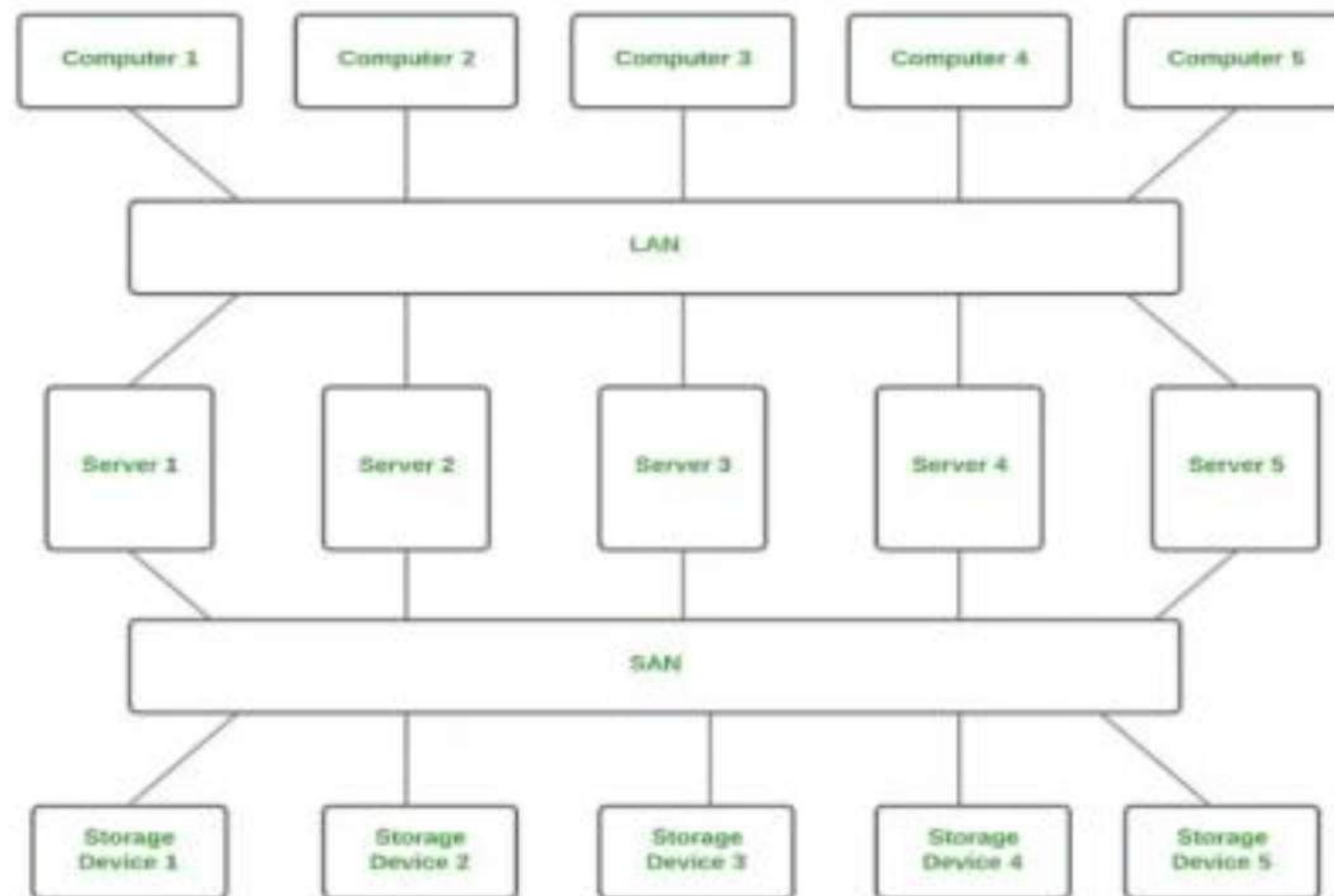
1. Wireless Local Area Network (WLAN) :

WLAN is a type of computer network that acts as a local area network but makes use of wireless network technology like Wi-Fi. This network doesn't allow devices to communicate over physical cables like in LAN but allows devices to communicate wirelessly. The most common example of WLAN is Wi-Fi.



2. System Area Network (SAN)

A System Area Network is designed to connect high-performance computers within a localized, high-speed environment, such as in data centers or supercomputing facilities. A SAN provides access to block-level data storage. Examples of SAN are a network of disks accessed by a network of servers.



Why storage area networks are important

Computer memory and local storage resources might not provide enough storage, storage protection, multiple-user access or speed and performance for enterprise applications. So, most organizations employ some form of SAN in addition to network-attached storage (NAS) for improved efficiency and better data management.

Traditionally, only a limited number of storage devices could attach to a server, limiting a network's storage capacity. However, a SAN introduces networking flexibility that enables one server, or many heterogeneous servers across multiple data centers, to share a common storage utility. The SAN eliminates the traditional dedicated connection between a network file server and storage—and the concept that the server effectively owns and manages the storage devices—eliminating bandwidth bottlenecks. A SAN eliminates single points of failure to enhance storage reliability and availability.

A SAN is also optimal for disaster recovery (DR) because a network might include many storage devices, including disk, magnetic tape and optical storage. The storage utility might also be located far from the servers that it uses.

1. Private Network

These are networks completely owned and managed by a single organization or individual. The owner controls who can connect, what they can do, and how data moves within the network. Since there is no outside access, they are highly secure and reliable, often protected by firewalls and strict policies.

Examples:

- A company's internal office network (Intranet)
- School or college campus networks
- Hospital systems storing patient data

2. Public Network

These are networks open to the general public and are usually provided by Internet Service Providers (ISPs) or businesses in public places. They offer basic or no authentication, making them less secure and vulnerable to attacks if not used carefully. Examples: Public Wi-Fi at cafes or airports, free hotspots in city centers, Internet (the largest public network).

3. Hybrid Network

A hybrid network blends private and public access, offering flexibility and role-based access control. Some parts are restricted (like internal systems), while others are open (like guest Wi-Fi). This setup is useful in environments where different users need different access levels. Example: A university network with private access for staff and students and limited access for guests.

Internetwork

An internetwork is a setup where two or more networks (like LANs or WANs) are interconnected using routers or

- gateways, allowing communication between them. This process is called internetworking.

There are two main types:

1. Intranet

An intranet is a private network used within an organization. It allows employees to access internal tools, documents, and systems securely. It's not available to the public, and typically uses private IP addresses. **Examples:** Company HR portals, Internal knowledge bases, Staff-only communication tools.

2. Extranet

An extranet is an extension of the intranet that provides limited, secure access to outsiders such as partners, vendors, or clients. It's used to collaborate across organizations while maintaining internal security. **Examples:** Supplier portals, Client dashboards for project updates, Partner login systems for shared services

Advantages of Computer Network

- **Central Storage of Data:** Files are stored on a central storage database which helps to easily access and available to everyone.
- **Connectivity:** A single connection can be routed to connect multiple computing devices.
- **Sharing of Files:** Files and data can be easily shared among multiple devices which helps in easily communicating among the organization.
- **Security through Authorization:** Computer Networking provides additional security and protection of information in the system.

Disadvantages of Computer Network

- **Virus and Malware:** A virus is a program that can infect other programs by modifying them. Viruses and Malware can corrupt the whole network.
- **High Cost of Setup:** The initial setup of Computer Networking is expensive because it consists of a lot of wires and cables along with the device.
- **loss of Information:** In case of a System Failure, might lead to some loss of data.
- **Management of Network:** Management of a Network is somehow complex for a person, it requires training for its proper use.

Network topology refers to the arrangement of different elements like nodes, links, or devices in a computer network. Common types of network topology include bus, star, ring, mesh, and tree topologies, each with its advantages and disadvantages. In this article, we will discuss different types of network topology in detail.

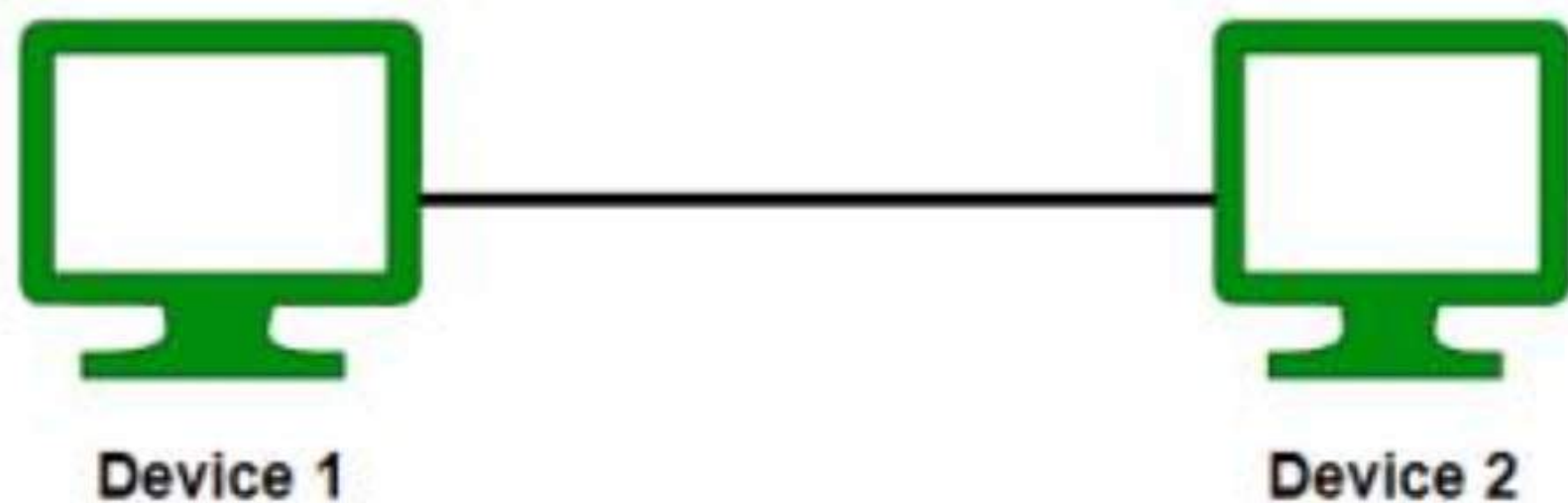
There are two major categories of Network Topology i.e. Physical Network topology and Logical Network Topology. Physical Network Topology refers to the actual structure of the physical medium for the transmission of data. Logical network Topology refers to the transmission of data between devices present in the network irrespective of the way devices are connected. The structure of the network is important for the proper functioning of the network. one must choose the most suitable topology as per their requirement.

Types of Network Topology

- Point to Point Topology
- Mesh Topology
- Star Topology
- Bus Topology
- Ring Topology
- Tree Topology
- Hybrid Topology

Point to Point Topology

Point-to-point topology is a type of topology that works on the functionality of the sender and receiver. It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver. Point-to-Point provides high bandwidth.



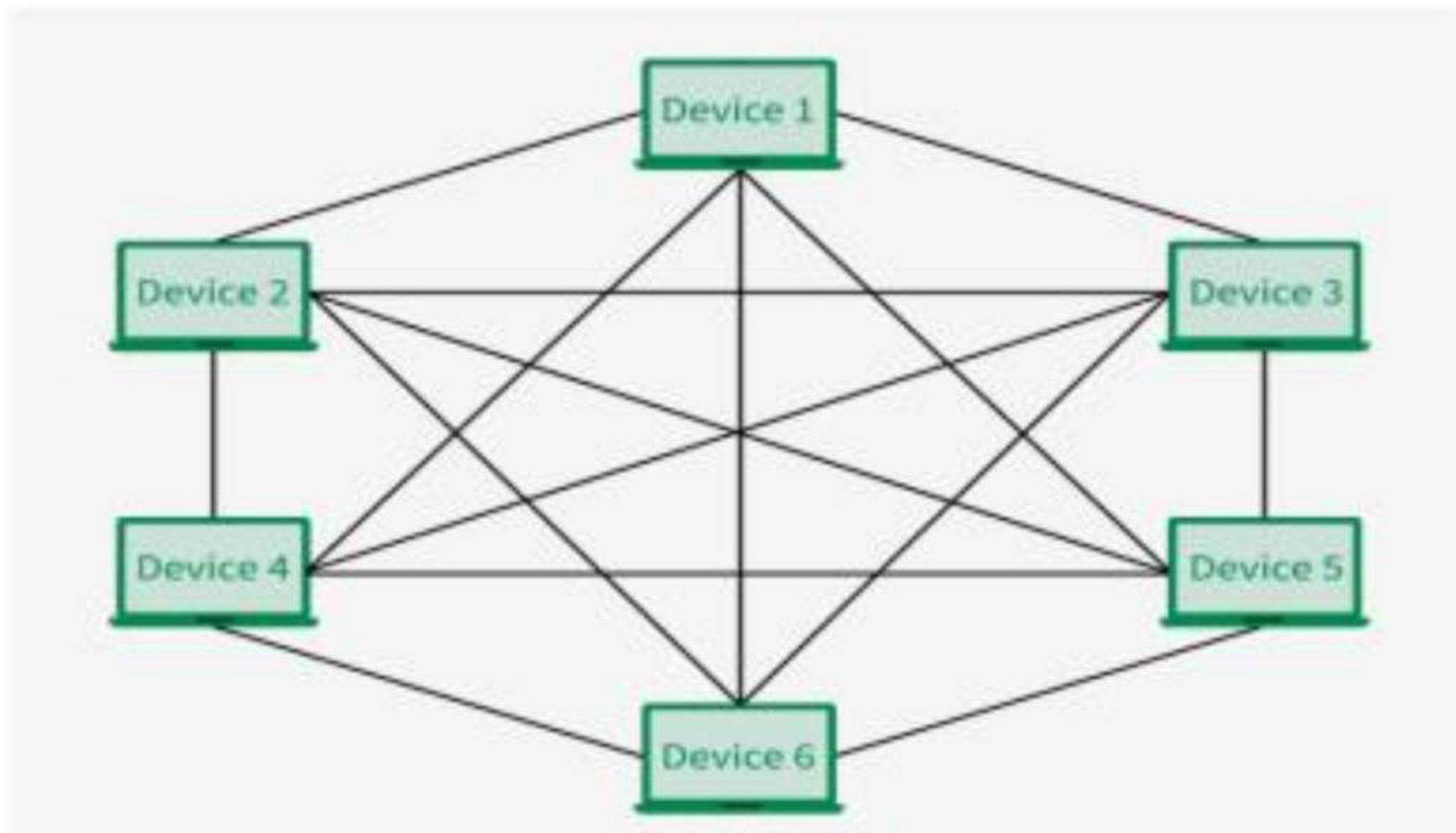
Point to Point Topology

Topic- Network Topology- MESH



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In a mesh topology, every device is connected to another device via a particular channel. Every device is connected to another via dedicated channels. These channels are known as links. In Mesh Topology, the protocols used are AHCP (Ad Hoc Configuration Protocols), DHCP (Dynamic Host Configuration Protocol), etc.



Mesh Topology

- Suppose, the N number of devices are connected with each other in a mesh topology, the total number of ports that are required by each device is $N-1$. In Figure, there are 6 devices connected to each other, hence the total number of ports required by each device is 5. The total number of ports required = $N * (N-1)$.
- Suppose, N number of devices are connected with each other in a mesh topology, then the total number of dedicated links required to connect them is ${}^N C_2$ i.e. $N(N-1)/2$. In Figure, there are 6 devices connected to each other, hence the total number of links required is $6*5/2 = 15$.

Advantages of Mesh Topology

- Communication is very fast between the nodes.
- Mesh Topology is robust.
- The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.
- Provides security and privacy.

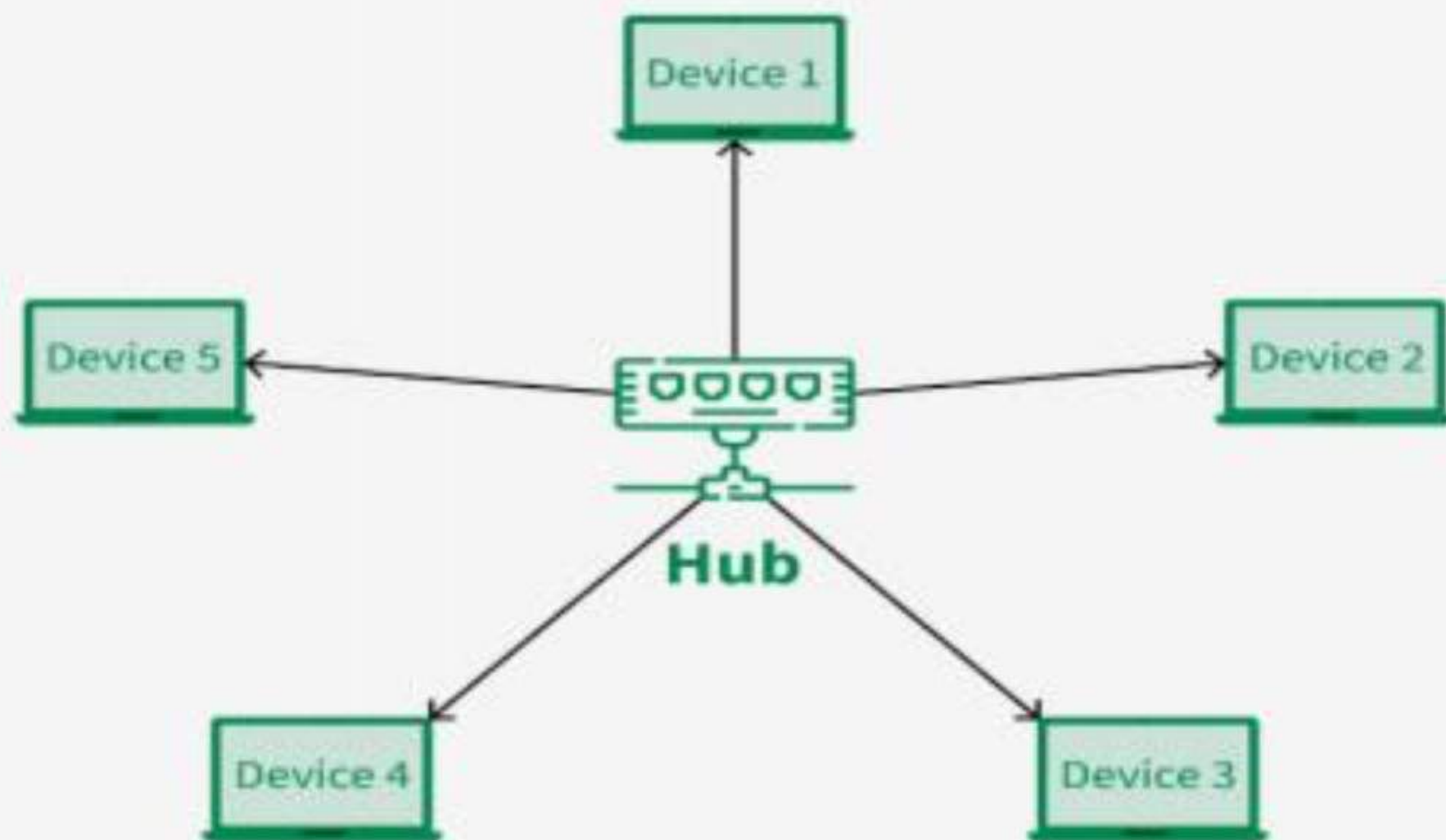
Disadvantages of Mesh Topology

- Installation and configuration are difficult.
- The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.
- The cost of maintenance is high.

A common example of mesh topology is the internet backbone, where various internet service providers are connected to each other via dedicated channels. This topology is also used in military communication systems and aircraft navigation systems.

Star Topology

In Star Topology, all the devices are connected to a single hub through a cable. This hub is the central node and all other nodes are connected to the central node. The hub can be passive in nature i.e., not an intelligent hub such as broadcasting devices, at the same time the hub can be intelligent known as an active hub. Active hubs have repeaters in them. Coaxial cables or RJ-45 cables are used to connect the computers. In Star Topology, many popular [Ethernet](#) LAN protocols are used as CD(Collision Detection), [CSMA](#) (Carrier Sense Multiple Access), etc.



Star Topology

Advantages of Star Topology

- If N devices are connected to each other in a star topology, then the number of cables required to connect them is N . So, it is easy to set up.
- Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N .
- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.
- Star topology is cost-effective as it uses inexpensive coaxial cable.

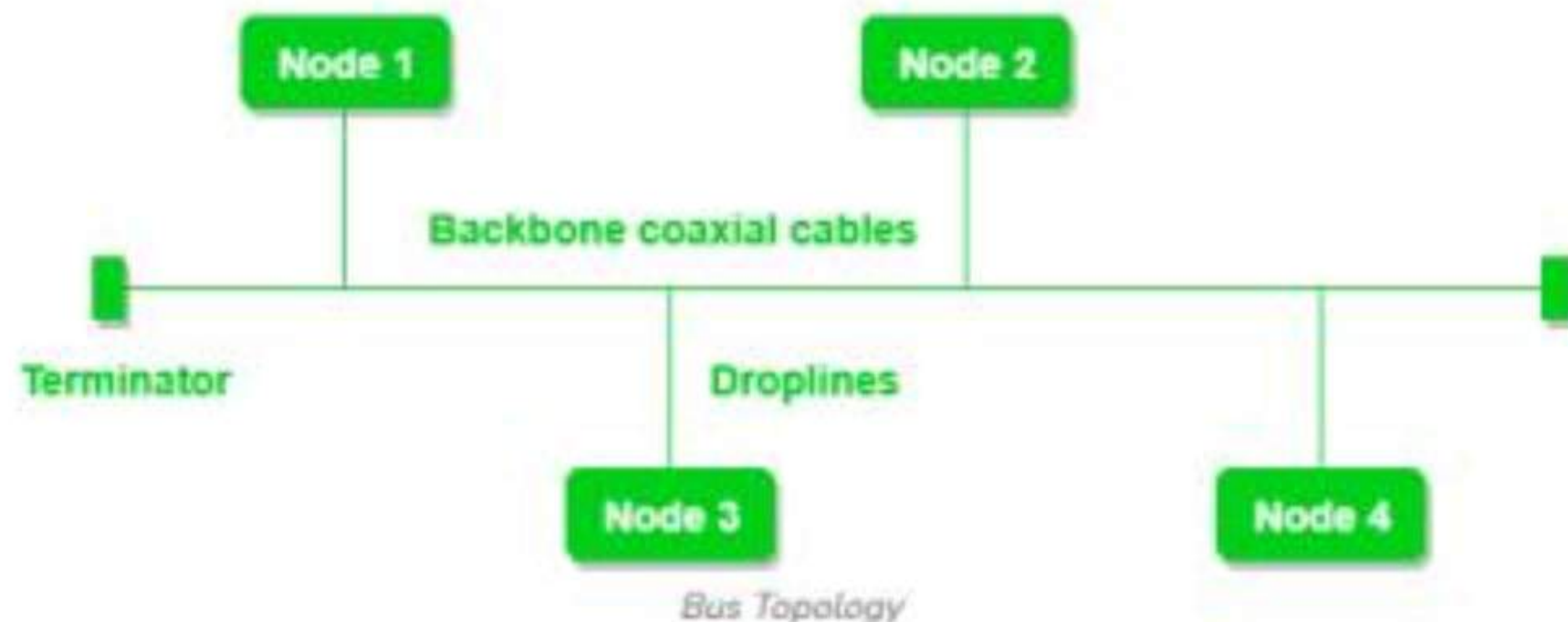
Disadvantages of Star Topology

- If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.
- The cost of installation is high.
- Performance is based on the single concentrator i.e. hub.

A common example of star topology is a **local area network (LAN)** in an office where all computers are connected to a central hub. This topology is also used in wireless networks where all devices are connected to a wireless access point.

Bus Topology

Bus Topology is a network type in which every computer and network device is connected to a single cable. It is bi-directional. It is a multi-point connection and a non-robust topology because if the backbone fails the topology crashes. In Bus Topology, various MAC (Media Access Control) protocols are followed by LAN ethernet connections like TDMA, Pure Aloha, CDMA, Slotted Aloha, etc.



Advantages of Bus Topology

- If N devices are connected to each other in a bus topology, then the number of cables required to connect them is 1, known as backbone cable, and N drop lines are required.
- Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.
- The cost of the cable is less compared to other topologies, but it is used to build small networks.
- Bus topology is familiar technology as installation and troubleshooting techniques are well known.
- CSMA is the most common method for this type of topology.

Disadvantages of Bus Topology

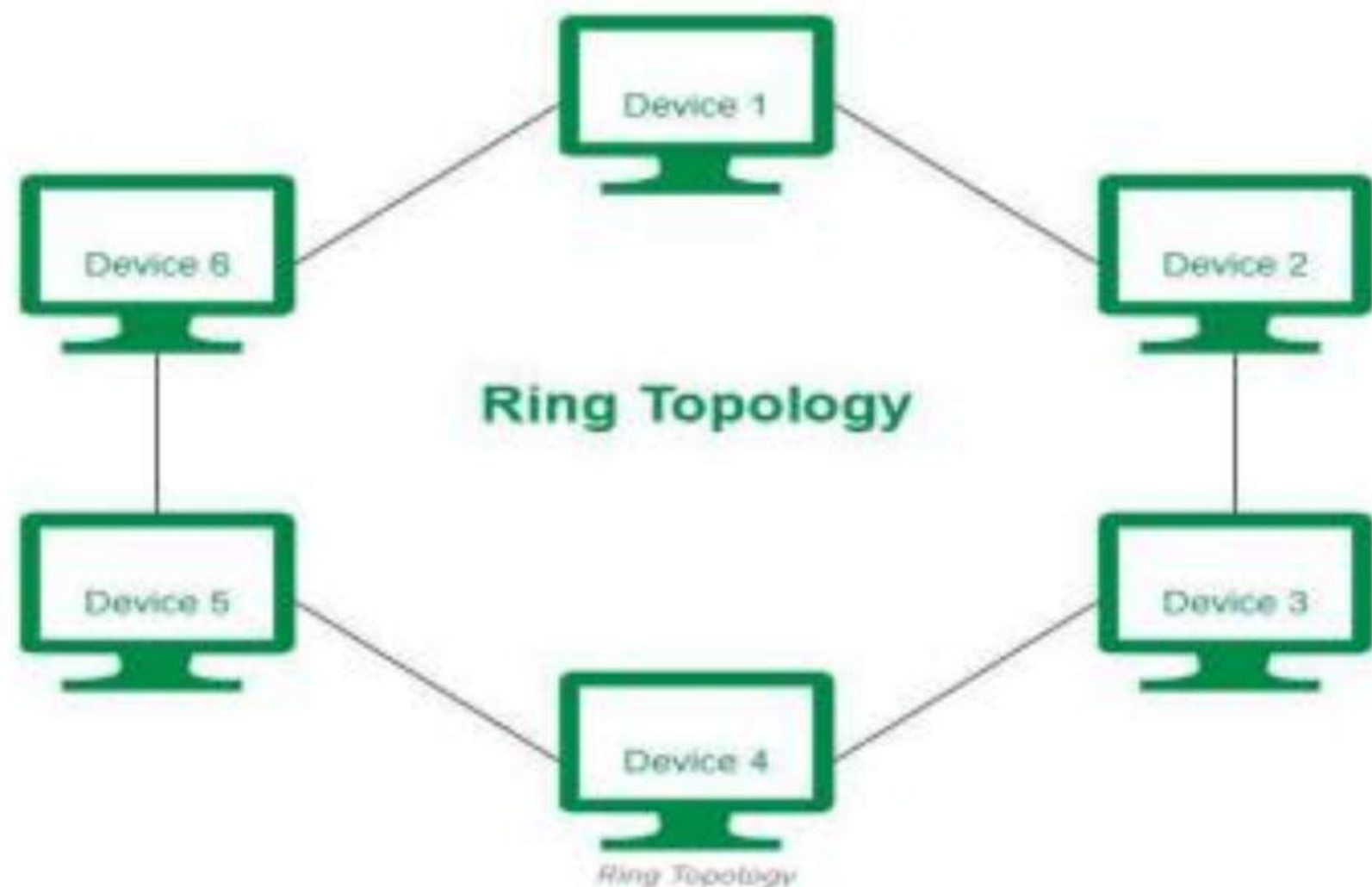
- A bus topology is quite simpler, but still, it requires a lot of cabling.
- If the common cable fails, then the whole system will crash down.
- If the network traffic is heavy, it increases collisions in the network. To avoid this, various protocols are used in the MAC layer known as Pure Aloha, Slotted Aloha, CSMA/CD, etc.
- Adding new devices to the network would slow down networks.
- Security is very low.

A common example of bus topology is the Ethernet LAN, where all devices are connected to a single coaxial cable or twisted pair cable. This topology is also used in cable television networks.

Ring Topology

In a Ring Topology, it forms a ring connecting devices with exactly two neighboring devices. A number of repeaters are used for Ring topology with a large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.

The data flows in one direction, i.e. it is unidirectional, but it can be made bidirectional by having 2 connections between each Network Node, it is called Dual Ring Topology. In-Ring Topology, the Token Ring Passing protocol is used by the workstations to transmit the data.



The most common access method of ring topology is token passing.

The most common access method of ring topology is token passing.

- **Token passing:** It is a network access method in which a token is passed from one node to another node.
- **Token:** It is a frame that circulates around the network.

Operations of Ring Topology

- One station is known as a **monitor** station which takes all the responsibility for performing the operations.
- To transmit the data, the station has to hold the token. After the transmission is done, the token is to be released for other stations to use.
- When no station is transmitting the data, then the token will circulate in the ring.
- There are two types of token release techniques: **Early token release** releases the token just after transmitting the data and **Delayed token release** releases the token after the acknowledgment is received from the receiver.

Advantages of Ring Topology

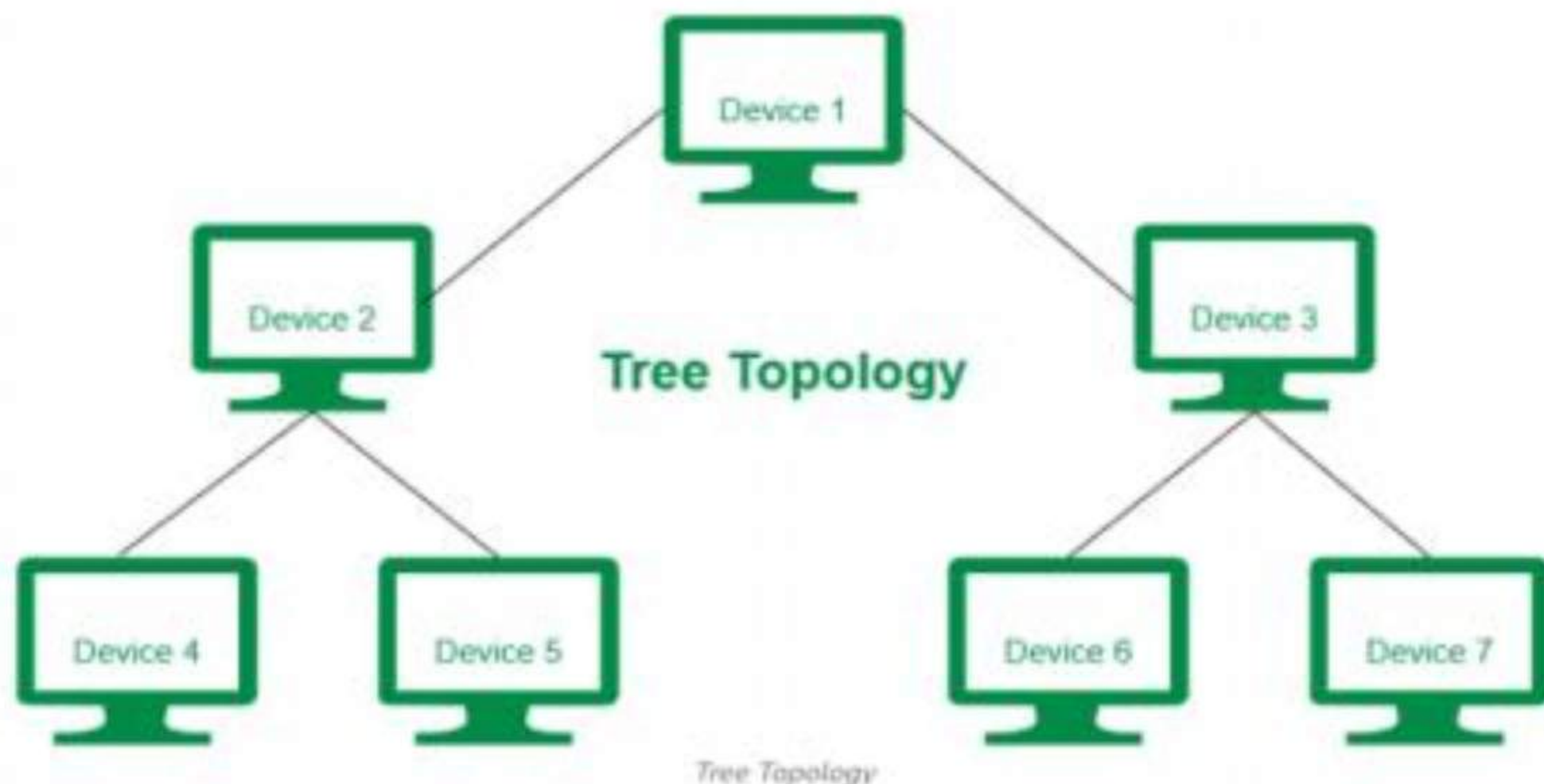
- The data transmission is high-speed.
- The possibility of collision is minimum in this type of topology.
- Cheap to install and expand.
- It is less costly than a star topology.

Disadvantages of Ring Topology

- The failure of a single node in the network can cause the entire network to fail.
- Troubleshooting is difficult in this topology.
- The addition of stations in between or the removal of stations can disturb the whole topology.
- Less secure.

Tree Topology

Tree topology is the variation of the Star topology. This topology has a hierarchical flow of data. In Tree Topology, protocols like [DHCP](#) and [SAC \(Standard Automatic Configuration\)](#) are used.



In tree topology, the various secondary hubs are connected to the central hub which contains the repeater. This data flow from top to bottom i.e. from the central hub to the secondary and then to the devices or from bottom to top i.e. devices to the secondary hub and then to the central hub. It is a [multi-point connection](#) and a non-robust topology because if the backbone fails the topology crashes.

Advantages of Tree Topology

- It allows more devices to be attached to a single central hub thus it decreases the distance that is traveled by the signal to come to the devices.
- It allows the network to get isolated and also prioritize from different computers.
- We can add **new devices to the existing network**.
- **Error detection and error correction** are very easy in a tree topology.

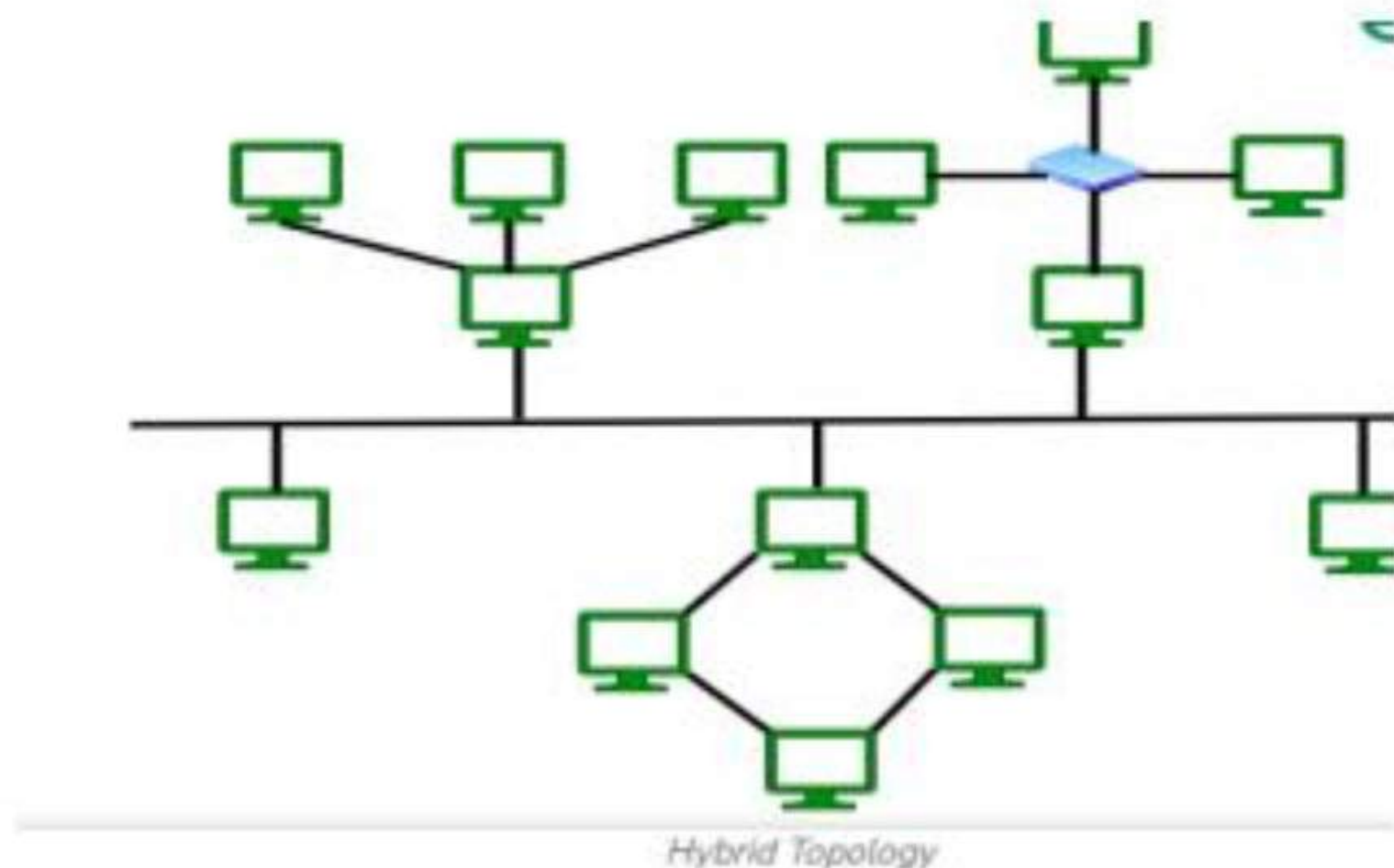
Disadvantages of Tree Topology

- If the central hub gets fails the entire system fails.
- The cost is high because of the cabling.
- If new devices are added, it becomes difficult to reconfigure.

A common example of a tree topology is the hierarchy in a large organization. At the top of the tree is the CEO, who is connected to the different departments or divisions (child nodes) of the company. Each department has its own hierarchy, with managers overseeing different teams (grandchild nodes). The team members (leaf nodes) are at the bottom of the hierarchy, connected to their respective managers and departments.

Hybrid Topology

Hybrid Topology is the combination of all the various types of topologies we have studied above. Hybrid Topology is used when the nodes are free to take any form. It means these can be individuals such as Ring or Star topology or can be a combination of various types of topologies seen above. Each individual topology uses the protocol that has been discussed earlier.



The above figure shows the structure of the Hybrid topology. As seen it contains a combination of all different types of networks.

Advantages of Hybrid Topology

- This topology is **very flexible** .
- The size of the network can be easily expanded by **adding new devices**.

Disadvantages of Hybrid Topology

- It is challenging to **design the architecture** of the Hybrid Network.
- **Hubs** used in this topology are **very expensive**.
- The infrastructure cost is very high as a hybrid network **requires a lot of cabling and network devices** .

A common example of a hybrid topology is a university campus network. The network may have a backbone of a star topology, with each building connected to the backbone through a switch or router. Within each building, there may be a bus or ring topology connecting the different rooms and offices. The wireless access points also create a mesh topology for wireless devices. This hybrid topology allows for efficient communication between different buildings while providing flexibility and redundancy within each building.

Why is Network Topology Important?

Network Topology is important because it defines how devices are connected and how they communicate in the network.

- Here are some points that defines why network topology is important.
 - **Network Performance:** Upon choosing the appropriate topology as per requirement, it helps in running the network easily and hence increases network performance.
 - **Network Reliability:** Some topologies like Star, Mesh are reliable as if one connection fails, they provide an alternative for that connection, hence it works as a backup.
 - **Network Expansion :** Choosing correct topology helps in easier expansion of Network as it helps in adding more devices to the network without disrupting the actual network.
 - **Network Security:** Network Topology helps in understanding how devices are connected and hence provides a better security to the network.

Conclusion

In conclusion, network topologies play a crucial role in determining the efficiency and reliability of a computer network. Each topology, whether it's bus, star, ring, mesh, or tree, offers unique benefits and potential drawbacks. By understanding these different arrangements, network designers can choose the most appropriate topology to meet the specific needs of their systems, ensuring optimal performance and connectivity.

THANK YOU